9.2.5 Limits at infinity

In the previous section we investigated infinite limits and vertical asymptotes. There, we let x approach a number and the result was that the values of y became arbitrarily large (very large positive or very large negative). In this section, we let x become arbitrarily large (positive or negative) and see what happens to y.

Let's begin by investigating the behaviour of $f : \mathbb{R} \to \mathbb{R}$ defined by

$f(x) = \dfrac{x^2 - 1}{x^2 + 1}$ as x becomes large.

We tabulate the values of this function

x	$f(x)$
0	-1
± 1	0
± 2	0.6000000
± 3	0.800000
± 4	0.882353
± 5	0.923077
± 10	0.980198
± 50	0.999200
± 100	0.999800
± 1000	0.999998

As x grows larger and larger (large positive or large negative) you can see that the values of $f(x)$ gets closer and closer to 1. In fact, it seems that we can make the values of $f(x)$ as close as to 1 by taking x sufficiently large. This situation is expressed symbolically by writing

$$\lim_{x \to \pm\infty} \frac{x^2 - 1}{x^2 + 1} = 1.$$

If we look at the graph :

Geometrically, (see Fig. 9.25) this situation also leads us to have

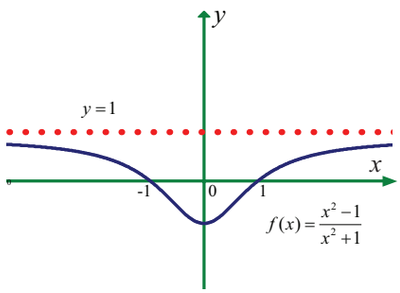


Fig. 9.25

Definition 9.6

The line $y = l$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x \to -\infty} f(x) = l \quad \text{or} \quad \lim_{x \to +\infty} f(x) = l.$$

Illustration 9.4

If $f : \mathbb{R} \to \left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$ is defined by $f(x) = \tan^{-1} x$,

find $\lim\limits_{x \to \infty} f(x)$ and $\lim\limits_{x \to -\infty} f(x)$.

Solution

If we look at the graph of $y = \tan^{-1} x$,

$$\lim_{x \to -\infty} \tan^{-1} x = -\frac{\pi}{2},$$

$$\lim_{x \to +\infty} \tan^{-1} x = \frac{\pi}{2}.$$

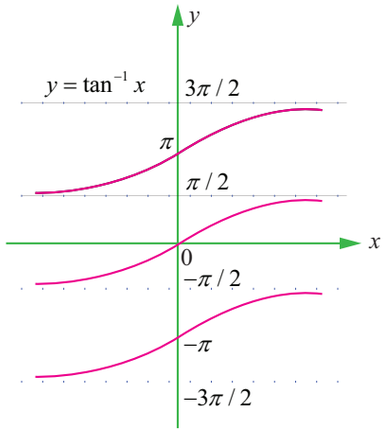


Fig. 9.26

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